

Nasim Mahmud Mishu

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RESEARCH INTEREST

My research interests focus on the intersection of **robotics and artificial intelligence**, with a particular interest on **swarm intelligence, AI-driven robotic systems, and computer vision**. I am also interested in CAD-based robotic design, motion planning for autonomous navigation, and the integration of perception and control mechanisms to enhance intelligent decision-making in robotic agents.

EDUCATION

- **North South University** Dhaka, Bangladesh
Bachelor of Science in Computer Science and Engineering May 2017 – December 2021
CGPA: 3.39/4.00 (87–89%)
Thesis Title: Autonomous Obstacle Avoiding Exploring Robot (AOAVER)

EXPERIENCE

- **Research Assistant** Dhaka, Bangladesh
NSU Intelligent Robotics (NIRO) Laboratory October 2024 – Present
Dept. of Electrical and Computer Engineering, North South University
Supervisor: [Dr. Shahnewaz Siddique](#)
Research Area: Swarm robotics with a focus on centralized and decentralized control systems
Research Focus:
 - Investigated swarm robotics architectures under centralized and decentralized control strategies.
 - Integrated machine learning techniques to enhance swarm behavior, coordination, and adaptability.
 - Designed experiments for evaluating swarm performance in dynamic environments.
- **Teaching Assistant** Dhaka, Bangladesh
Dept. of Electrical and Computer Engineering, North South University January 2025 – Present
Courses: Fall'25, Spring'25 (EE494/ CSE495A: Introduction to Robotics)
Summer'26, Summer'25 (EE494/ CSE495A, CSE543: Introduction to Robotics)
Primary Assignments:
 - Supported course delivery on robotics fundamentals (kinematics, motion planning, perception, control).
 - Assisted students through tutorials, homework evaluation, and project guidance.
 - Helped with exam invigilation and grading to maintain academic integrity.
- **Part-time Research Assistant** Dhaka, Bangladesh
Dept. of Electrical and Computer Engineering, North South University December 2023 – June 2024
Supervisor: [Dr. Shahnewaz Siddique](#)
Research Area: Resource-efficient visual Deep Reinforcement Learning (DRL) for autonomous agents and robots.
Research Focus:
 - Developed custom datasets for feature extraction in vision-based tasks.
 - Applied ML models for segmentation-based vision in autonomous agents.
 - Benchmarked approaches to improve resource efficiency in DRL models.
- **Founding Member** Dhaka, Bangladesh
NSU Ignite North South University September 2018 – February 2022
Primary Assignments:

- Designed and constructed remote-controlled and autonomous robots for competitions.
- Conducted hands-on workshops to train and inspire new robotics enthusiasts.

PROJECTS

- **NIRO Educational Bot** October 2024 – Present
 - Developed an educational robot at NSU Intelligent Robotics (NIRO) Lab to support hands-on learning and research.
 - Integrated core components (gyroscope, IR sensors, motors, WiFi) with Raspberry Pi and Arduino.
 - Designed a customizable 3D-printed chassis and PCB, enabling hardware extensions.
 - **Tools & technologies:** Raspberry Pi, Arduino, 3D Printing, PCB Designing
- **Project Hex** February 2025 – Present
 - Designed a hexapod robot with six legs (3 DOF each) for robust mobility on uneven terrain.
 - Integrated 18 high-torque bus servo motors, powered by LiPo battery for extended operation.
 - Implemented Arduino-based real-time control with modular support for Raspberry Pi and sensors.
 - **Tools & technologies:** Arduino, High-torque Servo Motors, LiPo Battery, Modular Robotics Design, 3D Printing
- **Autonomous Obstacle Avoiding Exploring Robot (AOAVER)** Summer 2021 — Spring 2022
 - Designed and developed an autonomous robot for search and rescue in disaster-prone areas.
 - Implemented obstacle avoidance and exploration algorithms for hazardous environments.
 - Integrated sensors and controllers for real-time perception and decision-making.
 - **Tools & technologies:** Python, YDLidar X4, Raspberry Pi 3 B+, Arduino UNO, Linux OS, ROS
- **Assist Seniors and Disabled with Hand Gesture Recognition** Spring 2020
 - Built a low-cost hand gesture recognition system to assist elderly and disabled individuals.
 - Enabled emergency calling through gesture-based interaction using a mobile app.
 - Applied computer vision for reliable gesture detection and recognition.
 - **Tools & technologies:** Python, OpenCV, Raspberry Pi 3 B+, Pi-camera, Firebase, Android Studio

PUBLICATIONS

Journal & Conference Papers

- N. M. Mishu, M. R. Hossain, M. S. A. Khan and R. Haque, “Accuracy-Oriented Deep Learning Based Hand Gesture Recognition Framework for the Elderly and Disabled,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox’s Bazar, Bangladesh, 2025, pp. 2509-2514. [\[DOI\]](#)
- Nelay, Asif Ahmed, Rafia Alif Bindu, Sazid Alam, Ridwanul Haque, Md. Saif Ahammod Khan, **Nasim Mahmud Mishu**, and Shahnewaz Siddique. “Alpha-N-V2: Shortest Path Finder Automated Delivery Robot with Obstacle Detection and Avoiding System.” *Vietnam Journal of Computer Science*, vol. 7, no. 4, 2020, pp. 373–389. World Scientific. [\[DOI\]](#)
- Nelay, Asif Ahmed, Rafia Alif Bindu, Sazid Alam, Ridwanul Haque, Md. Saif Ahammod Khan, **Nasim Mahmud Mishu**, and Shahnewaz Siddique. “Alpha-N: Shortest Path Finder Automated Delivery Robot.” *Asian Conference on Intelligent Information and Database Systems (ACIIDS 2020)*, edited by Ngoc Thanh Nguyen et al., vol. 12034, Springer, 2020, pp. 217–228. Lecture Notes in Computer Science. [\[DOI\]](#).

Special Proceedings

- Haque, Ridwanul, Md. Saif Ahammod Khan, **Nasim Mahmud Mishu**, Rahat Jahangir Rony, and Nova Ahmed. “Understanding the Healthcare Sector in Bangladesh: Experiences and Services during the COVID-19 Pandemic.” *6th Asian CHI Symposium 2022*, Apr. 2022, New Orleans, LA. North South University, Bangladesh. [\[ISBN\]](#)

REFERENCES

- **Dr. Shahnewaz Siddique**
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